

Unit 1 — Java Basics

Q1. Explain the complete execution process of a Java program from source code to output, covering the roles of the compiler, JVM, and class loader.

Q2. What are the scope and lifetime of a variable in Java? Explain local, instance, and class-level (static) variables with examples.

Q3. Explain the ternary operator and the pre/post increment and decrement operators with examples showing the difference in evaluation.

Q4. Compare the switch-case statement with the if-else ladder. What are the limitations of switch-case in older Java versions?

Q5. What are Wrapper Classes in Java? Explain Autoboxing and Unboxing with examples and their relevance in Collections.

Unit 2 — OOP Concepts

Q6. What is Encapsulation? How do access modifiers (public, private, protected, default) support encapsulation? Explain with a class example.

Q7. What is method overloading? How does Java resolve which overloaded method to call? How is it different from method overriding?

Q8. Describe the String class in Java. Why is String immutable? Explain the methods: length(), charAt(), substring(), toUpperCase(), and toLowerCase().

Q9. How does the Scanner class work for taking user input in Java? What is the difference between nextLine(), next(), and nextInt()?

Q10. Explain Garbage Collection in Java. What is the role of the finalize() method? Can a programmer force garbage collection?

Unit 3 — Inheritance, Interfaces & Packages

Q11. What is the final keyword in Java? Explain its use with variables, methods, and classes with examples for each case.

Q12. Explain the concept of an abstract class. What rules must be followed when using abstract classes and abstract methods?

Q13. What are default and static methods in interfaces (Java 8+)? How do they differ from regular interface methods and why were they introduced?

Q14. What are Java packages? Explain how to define a package, import it, and list the purpose of built-in packages like java.util and java.lang.

Unit 4 — Exception Handling & Collections

Q15. Explain the try-catch-finally block in Java. When is finally NOT executed? How do multiple catch blocks work (catch ordering rules)?

Q16. What is the difference between throw and throws in Java? Give an example of creating and throwing a custom exception.

Q17. Explain three common built-in exceptions: ArithmeticException, NullPointerException, and ArrayIndexOutOfBoundsException, including when they are thrown.

Q18. Describe the Collection hierarchy in Java. Explain the List and Set interfaces, and distinguish between ArrayList (with duplicates, ordered) and HashSet (no duplicates, unordered).

Unit 5 — Modern Java & AI Introduction

Q19. What is a Lambda Expression in Java 8? Explain the Stream API operations — filter(), map(), and forEach() — and how they enable functional-style programming.

Q20. What is the role of Java in Artificial Intelligence applications? Give a brief overview of Deeplearning4j (DL4J) and the TensorFlow Java API and their use cases.